

Dakai Kang

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Education

University of California, Davis

Computer Science, Ph.D. (M.S. awarded Dec 2025, en route to Ph.D.)

Sep 2022 – Jun 2026

Davis, CA

Research Team

Exploratory Systems Lab (expolab.org), advised by Prof. Mohammad Sadoghi (msadoghi@ucdavis.edu).

Dissertation

Towards High-Performance and Fair Byzantine Consensus: Concurrency, Speculation, Tail-Forking Resilience, and Transaction Ordering (ProQuest). Committee: Mohammad Sadoghi (Chair), Aditya Thakur, Amanda Raybuck.

Research Topics

- High-Performance Blockchain Infrastructure
- Fault-Tolerant Consensus Protocols; Transaction Scheduling; Parallel Execution

Teaching

- ECS165A, Databases, UC Davis, [Winter 2023](#), [2026](#) – Teaching Assistant
- ECS189F, Distributed Ledgers, UC Davis, [Fall 2023](#) – Teaching Assistant
- ECS265, Distributed Database Systems, UC Davis, [Fall 2024](#), [2025](#) – Teaching Assistant
- ECS188, Ethics in Computer Science, UC Davis, [Spring 2025](#) – Teaching Assistant

Awards

- GGCS Spring Quarter Fellowship Award, UC Davis (2023, 2024)
- GGCS Summer Ph.D. Fellowship (2023, 2024)

Zhejiang University

Software Engineering, Bachelor

Sep 2018 – July 2022

Hangzhou, China

Awards

- The Third Prize Scholarship of Zhejiang University (2019, 2020, 2021)
- Excellent Engineer Scholarship of Software Engineering Major (2022)

Work Experience

Sei Labs, Inc. (<https://www.sei.io>)

Research Scientist

Jun 2026 – Present

- Leading the Gas Repricing Project for the Sei blockchain to align on-chain gas costs with the actual resource consumption of the execution and storage layers.
- Contributing to a dual-path low-latency consensus project that reduces transaction commit latency by combining a fast optimistic path with a fallback path.

Mysten Labs, Inc. (<https://www.mystenlabs.com>)
Research Intern

Jun 2025 – Sep 2025
Remote

- Worked on a research project on transaction scheduling in Multi-proposer consensus to accelerate parallel transaction execution and mitigate congestion under high-contention workloads.

Hangzhou Zhuxing Information Technology Co., Ltd.
Fullstack Software Engineer Intern

Jul 2020 – Oct 2020
Hangzhou, China

- Developed a WeChat mini-program that enables the fourth official at football matches to record real-time match data.

Publications

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- [1] Xie, S., **Kang, D.**, Chen, J., Gupta, S., Hughes, D. P., & Sadoghi, M. (2027). Cassandra: Consensus with Partial Progress via Robust Partitionable View Synchronization. *Proceedings of the VLDB Endowment* (VLDB 2027). (*To appear.*) [arXiv:2607.02856](https://arxiv.org/abs/2607.02856)
 - [2] Dinh, T. T. A., **Kang, D.**, & Sadoghi, M. (2026). FEAT: Fair and Efficient Adversarial Transaction Ordering. In *Proceedings of the ACM Symposium on Principles of Distributed Computing (PODC '26)* (pp. 489–499). Association for Computing Machinery. <https://doi.org/10.1145/3796701.3815905>
 - [3] Xie, S., **Kang, D.**, Lyu, H., Niu, J., & Sadoghi, M. (2026). Fides: Scalable Censorship-Resistant DAG Consensus via Trusted Components. *Proceedings of the VLDB Endowment* (VLDB 2026). (*To appear.*) [arXiv:2501.01062](https://arxiv.org/abs/2501.01062)
 - [4] **Kang, D.**, Chen, J., Dinh, T. T. A., & Sadoghi, M. (2025). FairDAG: consensus fairness over multi-proposer causal design. *Proceedings of the VLDB Endowment*, 19(2), 265–278.
 - [5] Gupta, S., **Kang, D.**, Malkhi, D., & Sadoghi, M. (2025). Brief announcement: Carry the tail in consensus protocols. In *Proceedings of the 39th International Symposium on Distributed Computing (DISC 2025)* (Leibniz International Proceedings in Informatics [LIPIcs], Vol. 356, pp. 59:1–59:7). Schloss Dagstuhl – Leibniz-Zentrum für Informatik. <https://doi.org/10.4230/LIPIcs.DISC.2025.59>
 - [6] **Kang, D.**, Gupta, S., Malkhi, D., & Sadoghi, M. (2025). Hotstuff-1: Linear consensus with one-phase speculation. *Proceedings of the ACM on Management of Data*, 3(3), 1–29. (*Honorable Mention for Best Artifact*)
 - [7] **Kang, D.**, Rahnema, S., Hellings, J., & Sadoghi, M. (2024, May). Spotless: Concurrent rotational consensus made practical through rapid view synchronization. In *2024 IEEE 40th International Conference on Data Engineering (ICDE)* (pp. 1916–1929). IEEE.

Preprints

- [8] Ranchal-Pedrosa, A., **Kang, D.**, Giridharan, N., Malkhi, D., Sadoghi, M., & Marsh, B. (2026). Hermes: Low Tail-Latency Via Prefix Consensus. *arXiv preprint arXiv:2607.25916*.
- [9] Gupta, S., **Kang, D.**, Malkhi, D., & Sadoghi, M. (2025). Carry the Tail in Consensus Protocols. *arXiv preprint arXiv:2508.12173*.

Open-Source Projects

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- [1] ResilientDB (under Apache Incubation, <https://expolab.resilientdb.com>), **Architect**

- Github Repository: <https://github.com/apache/incubator-resilientdb>
- Implemented consensus protocols in ResilientDB, including PBFT, HotStuff-1 and FairDAG, etc.

Peer-Review Service

Program Committee Member

- International Symposium on Reliable Distributed Systems (**SRDS**) 2026

Invited Reviewer

- IEEE International Conference on Data Engineering (**ICDE**) (Demo Track) 2026
- IEEE Transactions on Information Forensics and Security (**T-IFS**) 2025
- IEEE Transactions on Dependable and Secure Computing (**TDSC**) 2025–2026
- ACM **SIGMOD** ARI (Availability & Reproducibility Initiative) 2025
- European Conference on Computer Systems (**EuroSys**) Shadow PC program 2026

External Reviewer

- ACM Conference on Computer and Communications Security (**CCS**) 2025
- International Conference on Very Large Data Bases (**VLDB**) 2023–2025
- ACM Special Interest Group on Management of Data (**SIGMOD**) 2024–2025
- IEEE International Conference on Data Engineering (**ICDE**) 2024
- ACM/IFIP International Middleware Conference (**Middleware**) 2024